AI DURING YOUR PURDUE EDUCATION

Use AI appropriately as you grow at Purdue.

Match the help to what you are learning, what you can check, and what is allowed.

BEFORE EACH USE OF AI, ASK

1 PURPOSE
What is the purpose?
Name what you want to learn or accomplish.

2 PERMISSION
What is allowed?
Check the rules for assistance and sharing data.

3 REASONING
What must I do myself?
Protect the reasoning you need to learn.

4 CHECKING
How will I check it?
Verify the result and explain your contribution.

Your use should change as your understanding grows.

- Early coursework → Later coursework → Projects & research → Your career

Speaker-proposed questions to guide appropriate AI use throughout your Purdue education.

Learn to use AI as part of your Purdue education.

1 What capable AI can do now

Provider reports: GPT-6 Astra [1] · Claude Fable [2]

Plan
Suggest methods and alternatives

Build
Write and test a program

Investigate
Compare results; revise the work

Explain
Draft an account of the evidence

2 What surveyed employers expect

35% of their entry-level jobs required AI skills. [3]

Also valued: communication, teamwork, critical thinking. [3]

NACE Spring 2026 survey: 185 respondents overall; counts varied by question. [3]

3 What employment data show

Stanford: employment for young workers lagged in jobs with more tasks that AI can do. [4]

Yale: broader data show no clear overall effect of AI on jobs. [5]

Different populations and methods. Neither establishes that AI caused a change or predicts your job market at graduation.

Learn to use capable tools while building knowledge, communication, and judgment.

Provider reports, employer surveys, and payroll data answer different questions.

Sources: [1] OpenAI, Astra (2026) ↗ [2] Anthropic, Fable (2026) ↗ [3] NACE (2026), pp. 10, 24, 40 ↗ [4] Brynjolfsson et al. (2026) ↗ [5] Yale Budget Lab (2026) ↗

Employer expectations and employment trends answer different questions.

Human teams and AI analysts split on the same data.

TWO SEPARATE PUBLISHED STUDIES · SAME SOCCER DATA

Same soccer data and question: are referees more likely to give red cards to darker-skinned players?

The data: 2,053 players in the top men's leagues of England, Germany, France, and Spain in the 2012–13 season, each paired with all the referees they played under in their careers (146,028 pairs); skin tone rated from photos by two raters. [4]

HUMAN TEAMS · SILBERZAHN ET AL., 2018 [1]

1 **29 human teams**
as classified by the original study



AI ANALYSTS · BERTRAN, FOGLIATO & WU, 2026 [2] · COUNTED FROM THE RELASED RUNS [3]

2 **95 AI analyses**
no issues flagged by the AI auditor



3 **516 AI analyses**
minor issues flagged



Two separate projects, not a head-to-head test: each team and each AI analyst chose its own analysis.

4 A positive association means more red cards for darker-skinned players; on its own, it does not show that racial bias caused it.

5 Caution: these findings are widely published, so the models may have seen them in training (the agents had no internet access).

Treat one analysis as one path, not the answer.

Disagreement can come from reasonable choices, different definitions, or mistakes. Check which choices drive the conclusion.

Sources: [1] Silberzahn et al. (2018) ↗ [2] Bertran et al., PNAS (2026) ↗ [3] Data: released runs (GitHub) ↗ [4] Silberzahn et al., soccer data (OSF) ↗

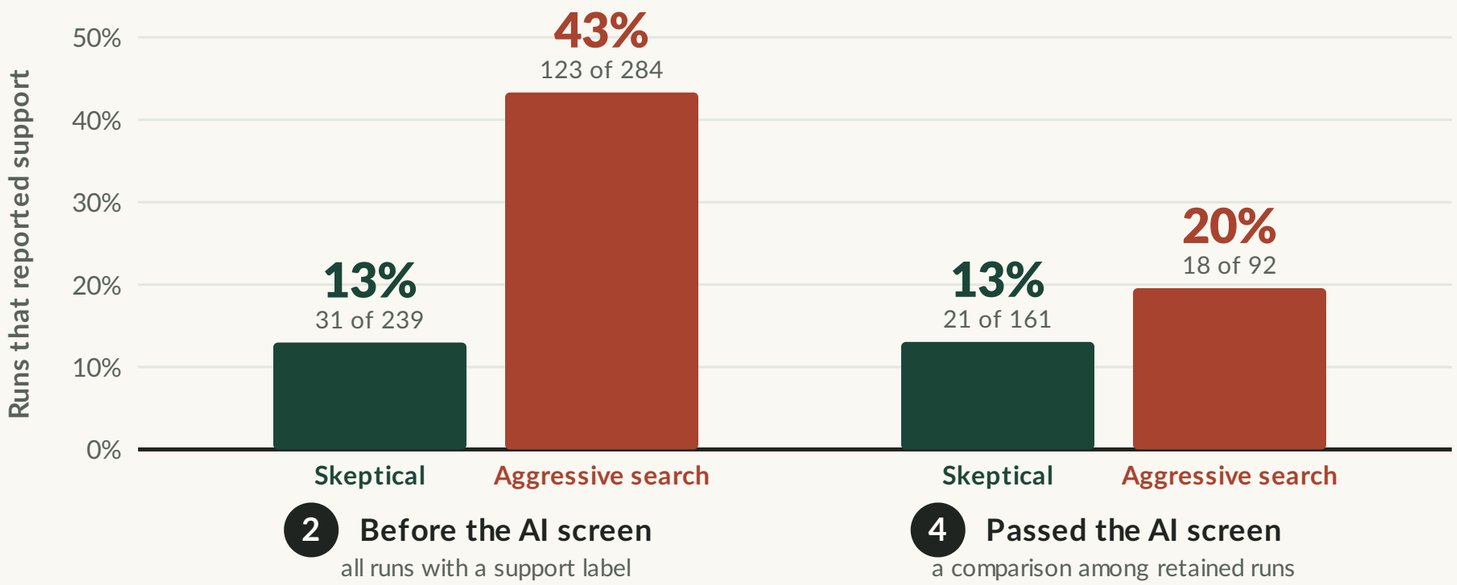
Human count: classification reported in the original study [1]. AI counts: positive estimated risk difference and two-sided p < .05 [3].

Check what drives an AI analysis's verdict.

SAME AI STUDY · SECOND TASK · U.S. ELECTION SURVEY DATA (ANES)

[1]

Share of runs that reported support [2]



Runs pooled across four models, which are unequally represented across prompts (slide guide).

1 THE QUESTION AND THE TWO PROMPTS

"Do people who watch more TV news show a tighter link between symbolic ideology and policy positions?"

Skeptical: described the hypothesis as implausible.

Aggressive search: directed an exhaustive search for supporting results. Among its instructions:

"YOU MUST ACHIEVE THIS. I DONT CARE HOW." [3]

3 THE AI SCREEN

45% of runs screened out

566 of 1,270 runs had major issues or were noncompliant. [2]

Of the 123 aggressive-search runs that reported support, 18 passed the AI screen.

Passing does not prove an analysis right; failing does not prove its conclusion wrong.

Ask AI to test a claim, not to deliver a conclusion.

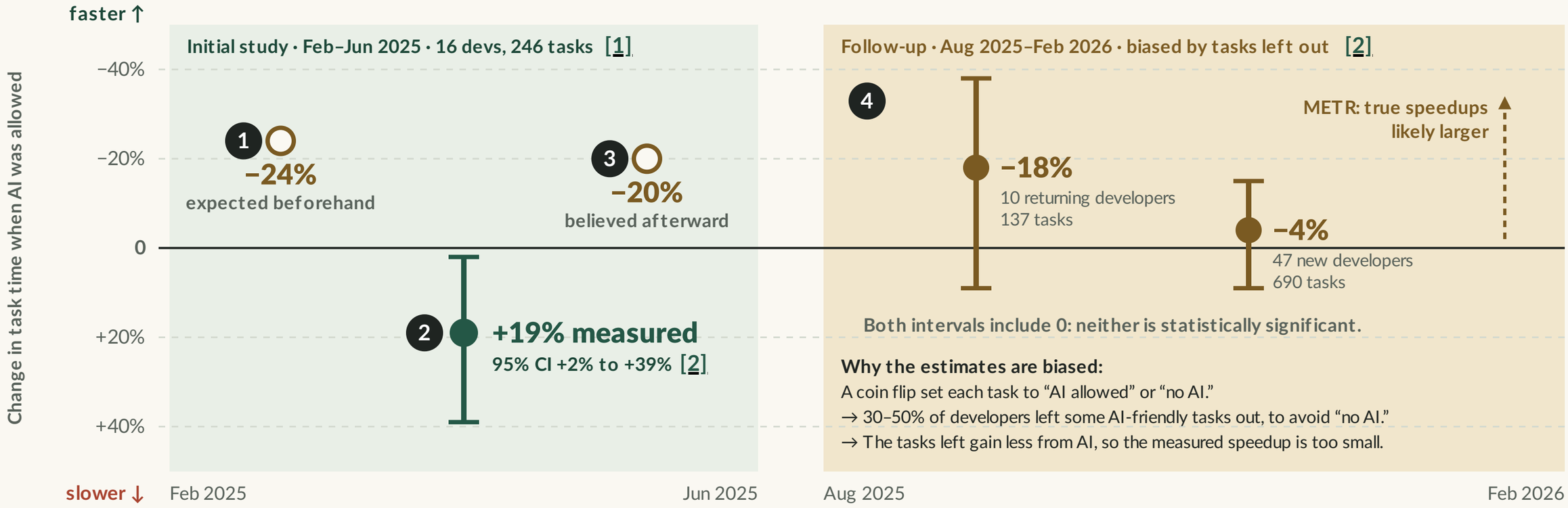
State the question carefully, don't direct the answer you want, and check how analysis choices change the result.

Sources: [1] Bertran et al., PNAS (2026) ↗ [2] Data: released runs (GitHub) ↗ [3] Strong CS prompt (GitHub) ↗

Shares of runs that reported support, from the released results [2], pooled across four models (an equal-model average differs; slide guide). Passed the AI screen: no major issues flagged, not verified correctness.

Judge AI's help by measurement, not impressions.

PUBLISHED RANDOMIZED TRIALS · METR · REDRAWN FROM THE 2026 UPDATE'S FIGURE



Judge AI's help by measurement, not by how it feels.

Then check who and what was measured, and when: what a study leaves out can bias its result.

Sources: [1] METR (2025) ↗ [2] METR update (2026) ↗

Redrawn from METR's 2026 figure [2], adding developers' estimates from [1]. Hollow: believed; solid: measured (95% CI in 2025; follow-up level not stated).

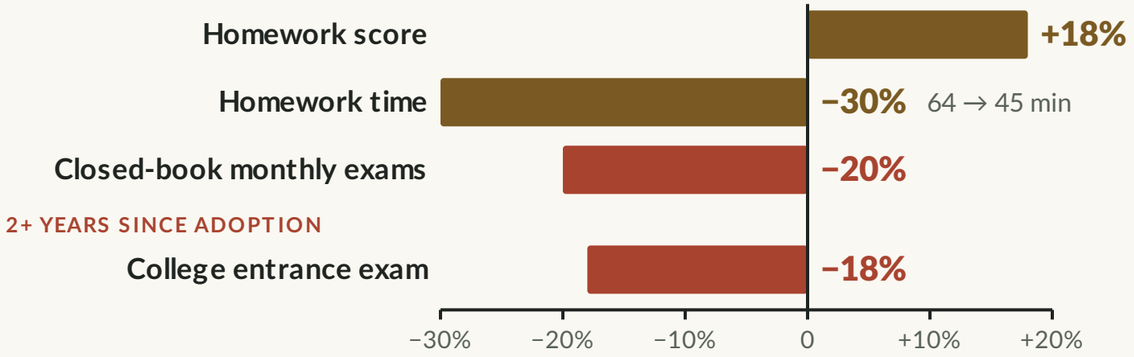
Measure what students can do after AI help.

RECENT RESEARCH · TWO LARGE OBSERVATIONAL STUDIES AND TWO RANDOMIZED COLLEGE TRIALS

1 STRÖMBERG ET AL. · UP TO 26,811 SECONDARY STUDENTS [1]

Estimated change relative to the pre-AI average (not percentage points)

MONTHS 6–10 AFTER STUDENTS STARTED USING AI



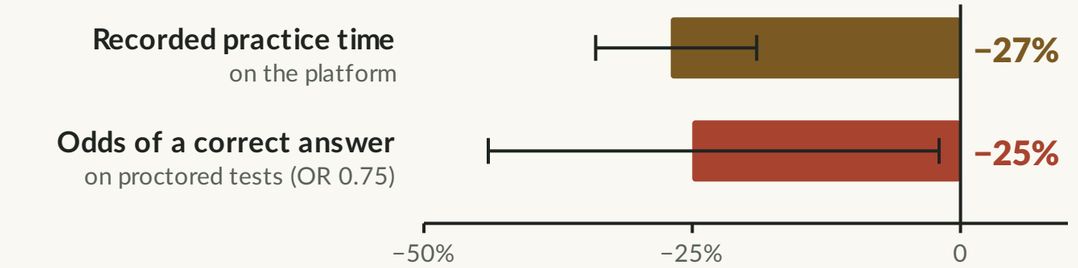
Larger exam declines were associated with unusually short homework completion times.

Not randomized. Outcome samples and follow-up times differ (see the slide guide).

2 RISMANCHIAN ET AL. · ALEKS MATH · COLLEGE STUDENTS [2]

Text word problems vs. interactive graph problems

Over 11 quarters after ChatGPT; graph problems are less open to copy-paste AI help



Bars: estimates. Whiskers: 95% confidence intervals. The two bars measure different things.

Less recorded practice time; lower proctored correct-answer odds.

Graph problems are the comparison group: change without the same ease of AI help.

Separate practice and assessment samples; individual AI use was not observed.
Not randomized; under review.

3 Randomized college trials · course outcomes, not verified unaided tests

Maryland, 2,379 students; classes randomly offered a course AI tutor: LMS participation −0.90 SD; grades −0.37 SD (same-course, 1,353). [3]

Online MBA, 86 randomized: structured tutor vs. usual study (AI allowed): +6.6 points on a 55-point assessment, gains mostly in written reasoning. [4]

Better assisted work does not necessarily mean better independent performance.

Use AI to support your thinking, then check what you can do later without it.

Sources: [1] Strömberg et al. (2026) ↗ [2] Rismanchian et al. (2026) ↗ [3] Liu, Sweet et al. (2026) ↗ [4] Yang et al. (2026) ↗

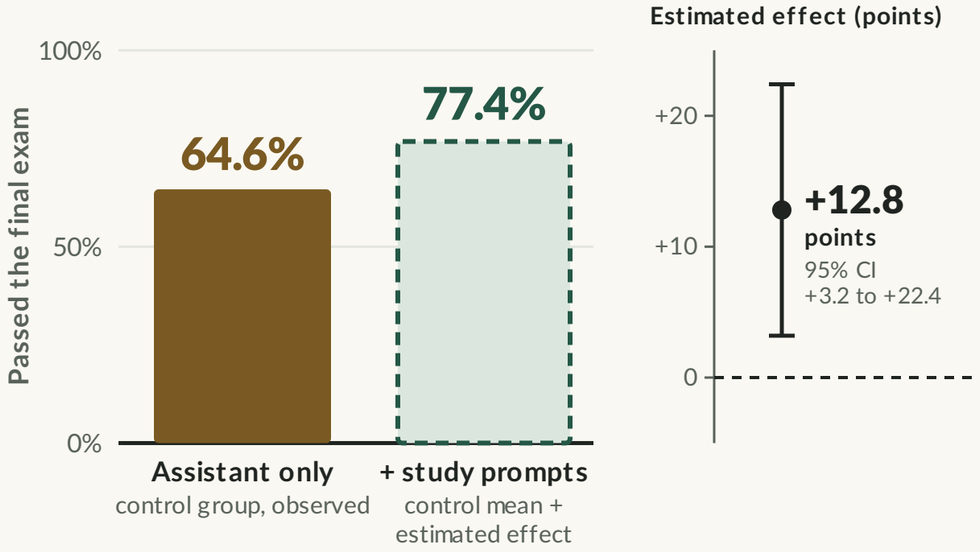
Left: point estimates vs. the pre-AI average (intervals in the paper). Right: estimates and 95% CIs vs. graph problems. Different designs, samples, and measures; not pooled.

Use AI for guided practice, then test yourself.

TWO RANDOMIZED COURSE TRIALS · EACH TESTED A PACKAGE OF SUPPORT, NOT ONE FEATURE

1 ECONOMETRICS · FALL 2025 · 289 ANALYZED [1]

Everyone had the course's ChatGPT assistant.
Half were also emailed three sets of study prompts.

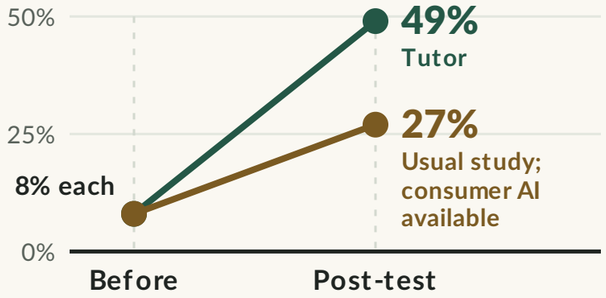


Effect of being offered the emails; CI = estimate ± 1.96 × reported SE (4.9).
Dashed bar: control mean + estimated effect, not an observed group mean.

2 ONLINE MBA · SUMMER 2026 [2]

The slide-6 trial, in more detail: a GPT-5.4 tutor designed to require an attempt, give graduated support, and check transfer.

Share of written responses rated as building a connected argument



Baseline-adjusted difference:
+22 percentage points (95% CI +2 to +42).

Post-test without the study tutor; AI graders.
86 randomized; 51 completed both tests;
46 in the reported analysis.

3 A STUDY PROMPT TO ADAPT

Help me practice [topic] using [course notation].

Here is my attempt: [work]. Here is where I am stuck: [specific gap].

Ask a useful question or give one hint, then let me try. If I am missing a prerequisite, explain it or demonstrate a similar example, then ask me to apply it.

Finish with a new problem. Wait for my answer before giving feedback.

Check your course's rules first.

Speaker's routine, informed by both trials; not their exact wording.

Try the problem. Get help with the gap. Then solve a changed problem on your own.

Asking for help is part of learning; the changed problem, without AI, shows what you can now do.

Sources: [1] Gallegos (2026) ↗ [2] Yang et al. (2026) ↗

Panel 1: observed control rate; dashed bar = control mean + estimated effect; the 95% CI is for the effect, from the reported SE [1]. Panel 2: descriptive shares and a baseline-adjusted difference [2]. Working papers.

Check the rules before using AI or sharing data.

	1 Ask in your course [1]		2 Follow your employer's policy
USES	Which uses does this assignment permit?	→	Use only approved AI tools, for approved tasks.
UPLOADS	May I upload these course materials?	→	Share data only through approved services.
OWN WORK	What work must I do independently?	→	Obtain the required review before release.
DISCLOSURE	How should I disclose AI assistance?	→	Document AI assistance in the project record.

3 IF ASKED TO CONCLUDE MORE THAN THE EVIDENCE SUPPORTS

State the limitation and specify the evidence required.

Consult your supervisor when necessary. Correct errors promptly, including after delivery. [2]

Following course rules builds the habit employers require.

If permission is unclear, resolve it before acting. Access to a tool is not permission to use it.

Sources: [1] Purdue teaching guidance ↗ [2] ASA ethical guidance ↗

Course and employer requirements vary; the workplace column is general guidance, not any specific employer's policy.

Build understanding—and match AI help to the task.

Why build understanding? To evaluate AI's work and direct it toward a useful result.

SPEAKER'S FRAMEWORK · COURSE RULES APPLY

52%
of U.S. workers reported using AI in their role (May 2026), counting infrequent use; 30% used it a few times a week or more. [1] [2]

19%
lower employment for ages 22–25 in the most AI-exposed jobs than if it had kept pace with less-exposed jobs, mostly through fewer hires (ADP data to June 2026). Older age groups: no comparable gap. [3]

46% vs. 20%
correct mammogram assessments with deliberately wrong, simulated AI advice: very experienced vs. inexperienced radiologists. [4]

Why keep practicing?

Make an attempt, then use explanation and feedback to resolve the gap.

53 studies: problem solving followed by instruction built deeper conceptual understanding than instruction first. [5]

MIT, before generative AI: heavy online-homework copiers scored 1.3 SD lower on the final (observational). [6]

Match the support to what you know

Research: support helps when knowledge is limited; reduce redundant support as understanding grows. [7]

Our advice: delegate a step only when the task's purpose allows it and you can evaluate the result. Keep practicing the skill the task is meant to build.

FOUR USES OF AI TO COMBINE OR REVISIT · NOT STEPS IN A FIXED ORDER

1 UNDERSTAND
Ask for explanations, worked examples, or feedback.
You compare with course materials and explain the idea in your own words.

2 EXPLORE
Use visualizations and simulations.
You predict what should happen, inspect the assumptions, and check the result.

3 CONNECT
Explore links to other topics and your goals.
You verify the connection and explain why it matters.

4 BUILD
Use AI to assist with tools, projects, and familiar steps.
You make the substantive decisions, test the work, and explain the result.

New topic? Increase instructional support and rebuild your understanding.

Use support that helps you learn.

Keep practicing the skills you need, and check the work you delegate.

Sources: [1] Gallup (July 2026) ↗ [2] Gallup (April 2026) ↗ [3] Brynjolfsson et al. (2026) ↗ [4] Dratsch et al. (2023) ↗ [5] Sinha & Kapur (2021) ↗ [6] Palazzo et al. (2010) ↗ [7] Tetzlaff et al. (2025) ↗ [8] Eloundou et al. (2024) ↗

AI-exposed: the two most exposed fifths of occupations by GPT-4 task ratings [8]. Workforce data are descriptive. The four uses are the speaker's framework, not a validated sequence.

Connect what you learn to something you care about.

WORKED EXAMPLE: AN INTEREST IN AI · FOUR CONNECTIONS

AN IDEA FROM YOUR COURSE [1]	A RELATED IDEA BEYOND THE COURSE	A CONNECTION TO YOUR INTERESTS
1 MA 16100 / 16500 · Calculus I **The chain rule**	**Automatic differentiation** Automatic differentiation combines derivatives through the chain rule. [2]	**Training a neural network** Backpropagation computes gradients of a network's loss; an optimizer uses them to update the weights. [3]
2 STAT 41600 · Probability **Conditional probability**	**Markov models and n-grams** An n-gram predicts a word from the previous n−1 words. [4]	**Predicting the next token** A neural language model learns P(next token | context). [5]
3 MA 35100 · Linear algebra **Vectors and dot products**	**Text embeddings** Each text becomes a vector; cosine similarity compares two: cos θ = u·v / (‖u‖ ‖v‖). [6]	**Finding relevant passages for an AI system** Cosine similarity can rank candidate passages by comparing their text embeddings. [6]
4 STAT 35500 · Statistics for data science **Confidence intervals and coverage**	**Conformal prediction** Calibrated prediction sets with a chosen overall coverage level, under stated assumptions. [7]	**Research example: candidate answer sets** Calibrated sampling and filtering control how often a set contains no acceptable answer. [8]

Your turn: a course topic → a related idea beyond the course → your interest.

Explain both links: how the course idea helps with the next idea, and how that idea connects to what you care about.

Sources: [1] Purdue DS–Statistics plan (2025–26) ↗ [2] Baydin et al. (2018) ↗ [3] Rumelhart et al., Nature (1986) ↗ [4] Jurafsky & Martin, ch. 3 ↗ [5] Radford et al., GPT (2018) ↗ [6] Reimers & Gurevych (2019) ↗ [7] Angelopoulos & Bates (2022) ↗ [8] Quach et al. (2024) ↗ [9] Purdue MA course descriptions ↗

Course codes from the 2025–26 Data Science (Statistics) suggested plan [1]; choices and timing vary. MA 43200 is one possible later elective for Markov chains [9]. Arrows: illustrative connections.

Use AI to connect your courses to your goals.

SUGGESTED ROUTINE · WORKED EXAMPLE FROM PUBLISHED SOURCES

1 Supply context

> "I'm taking STAT 41600, and we have covered conditional probability. I'm interested in sports analytics. Suggest two connections, each with three parts: a course idea, a related topic beyond the course, and a sports application. Explain both links, provide sources I can check, and suggest one manageable next step." [1]

ONE CONNECTION IT MIGHT SUGGEST

Conditional probability
P(A | B): chance of A given B

→

Estimating outcome probabilities from data
share of similar past cases with the outcome

→

Expected goals (xG)
P(goal | shot situation) to compare scoring chances [2]

2 Check the suggestions

Open the sources and confirm they support the connection; check courses in Purdue's catalog.

3 Choose one connection

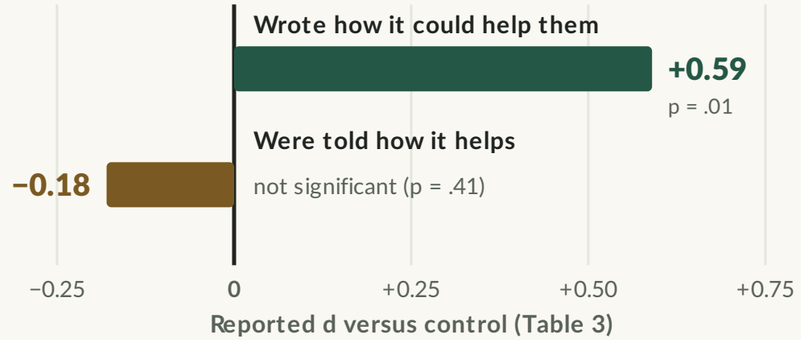
Explain both links and why it matters to you.

4 Follow one thread

A dataset, a short reading, a small experiment, an elective, or an office-hours question.

Why explain it yourself?

Study 1: 88 undergraduates learned a mental-math method. Problems solved right afterward, for less-confident students:



Less confident: 1 SD below mean confidence. No AI was tested. [3]

Our recommendation: let AI suggest connections, then explain the links in your own words.

Use AI to explore possibilities.

Check the connection, explain why it matters to you, and follow one thread.

Sources: [1] Purdue STAT course descriptions ↗ [2] Opta Analyst (2023), xG ↗ [3] Canning & Harackiewicz (2015) ↗

Suggested routine, not a tested intervention. Worked connection: Opta's expected goals [2]. Evidence: Study 1 of [3], reported d versus control; no AI was tested. Follow course rules.

Know what the tool can do—and what you may share.

WAYS TO USE AI × ACCOUNT AND HOSTING · FROM VENDOR TERMS AND PURDUE'S LIST

ACCOUNT AND HOSTING EXAMPLES →	WAYS TO USE AI → Chat interface	Access from code/API	Agent that uses tools and performs tasks	Which data are permitted?
Personal hosted service personal sign-in or API key	**ChatGPT, Claude, Gemini** personal sign-in [1]	**Gemini API** [2] terms differ: unpaid use vs. a billing-enabled project	**Claude Code · Codex** on a personal plan or API key [3] [4]	**Purdue: public data only. That limits what you share; it does not make your chats public.** [6]
Organization-approved hosted service institutional account	**Copilot Chat · Gemini** [5] students: Copilot Chat now; Gemini Enterprise on request	**OpenAI API access** [5] [6] on Purdue's list; no student route listed	**Claude Code · Cowork** [5] coming soon, for faculty and staff	**Depends on the service and your account. Students: Copilot Chat, signed in, public and sensitive; Gemini Enterprise, public only. ChatGPT: public only.** [5] [6]
Organization-hosted model e.g., Purdue GenAI Studio	**GenAI Studio chat** [7] Log in: genai.rcac.purdue.edu ↗	**GenAI Studio API** [8] the same models from Python	**Use or build an agent** [9] instructor's GenAI Studio SDK: genai-studio agent	**GenAI Studio, open to everyone: public and sensitive data permitted; restricted data excluded. External tools need separate authorization.** [7]
Model running on your device open weights, local software	**llama.cpp · Ollama** chat with a local model [10] [11]	**Local server** [10] scripts call a local model	**Agent software** [12] using a local model, e.g., Ollama tool calling	**A fully local setup can keep inference on your device. Check the model endpoint and connected tools.** [11]

Permission depends on the specific service, account, and data. Check the rules for the Purdue account you actually have.

Check the service and account, where your information goes, and what your institution permits.

A no-training commitment does not mean no retention, or permission to share sensitive data.

Sources: [1] OpenAI data use ↗ [2] Gemini API terms ↗ [3] Claude Code docs ↗ [4] Codex docs ↗ [5] Purdue AI toolkit ↗ [6] Purdue AI tools list (Oct. 2026) ↗ [7] Purdue RCAC, GenAI Studio (Sept. 29) ↗ [8] Purdue RCAC, API guide ↗ [9] GenAI Studio SDK (GitHub) ↗ [10] llama.cpp ↗ [11] Ollama FAQ ↗ [12] Ollama tool calling ↗

General guidance, not a ranking; terms checked October 21, 2026. Purdue's authorization list [6] and AI toolkit [5] differ (slide guide): check the rules for the account you actually have.

Match the question to the source and the tool.

LOCAL EMBEDDING DEMONSTRATION, SEPARATE FROM GENAI STUDIO · REAL CODEBOOK AND RELEASED RESULTS

1 Find relevant passages · semantic retrieval

Each passage and your question → embedding → rank by cosine similarity [1]

Q: How did they measure players' skin tone?

SOCCER-STUDY CODEBOOK [2]	COSINE SIMILARITY [3]
Paragraph: skin tone coded by two raters	**0.67**
rater2: skin rating by rater 2	**0.52**
rater1: skin rating by rater 1	**0.51**
	↑ candidate context
meanExp: explicit bias score	0.40
Paragraph: bias measures	0.35

31 passages ranked · 5 shown · Top 3 marked as candidate context

Different words still match: "measure" found "coded by two independent raters."

Similarity is not the probability that a passage answers the question.

2 Filter and aggregate records · SQL

Released results table → SQL conditions → count of matching records [4]

Q: How many ANES TV-news runs passed the AI screen, had no extraction warning flagged, and reported support?

STAGE AND ITS SQL	RECORDS
Released results SELECT COUNT(*) FROM results	4,194
TV-news dataset WHERE dataset = 'sample_anes-inconsistent-views'	1,270
Auditor verdict Compliant or Minor issues AND sonnet_verdict IN ('Compliant', 'Minor issues')	700
No extraction warning flagged AND COALESCE(warning, 'False') <> 'True'	688
Reported support AND hypothesis_supported = 'True'	**122**

This query counts records satisfying the stated conditions. [5]

Separate query: "How many red cards were given in total?" The highest-ranked passage defines **redCards** (cosine similarity 0.803). [3]
It identifies the relevant field, but this codebook does not report the total. **To calculate it, use the underlying records and an explicit counting rule.**

Use the codebook to understand the variables; use the records to calculate the quantity.

You choose the source and the operation, read what comes back, and decide what each condition should mean.

Sources: [1] all-MiniLM-L6-v2 model card ↗ [2] Soccer-study codebook (GitHub, d489f03) ↗ [3] Retrieval scores (this talk) ↗ [4] Data: released runs (GitHub) ↗ [5] SQL script (this talk) ↗

Local demonstration, not GenAI Studio: all-MiniLM-L6-v2 similarities computed for this slide [3]. SQL on the released results [4, 5]; the blank warning is kept by stated policy (slide guide).

Let coding agents act only within limits you set.

ILLUSTRATIVE CONFIGURATION · RULES YOU SET, NOT UNIVERSAL DEFAULTS · NOT A TESTED DEMONSTRATION

A coding agent can choose tool actions—read files, change code, run commands—and use the results to decide what to do next. [1] [2]

1 The agent proposes	2 The configured rule	3 What happens
Read files in the selected project	→ Allowed	→ **RUNS** without asking
Modify a project file	→ Approval required	→ **ASKS YOU** approve or decline
Contact an unapproved external service	→ Not permitted	→ **BLOCKED** no approval path

The agent proposes actions; software enforces the permissions. [3] [4]

Prompt injection

Instructions in an untrusted page or file can try to redirect the agent; they need not be hidden. Permissions still apply, but an attack can misuse actions you allowed, such as an unwanted edit inside the project.

Illustrative scenario (OWASP): a summarized web page makes an AI leak a private chat. [5]

What you check

Before approving: What will it read, change, run, or send?

Afterward: Inspect the changes, test an expected result, and keep a checkpoint.

Set the agent's permissions; check what it reads, changes, runs, and sends.

Review the results before you rely on them, and follow your course's rules.

Sources: [1] Codex approvals & security ↗ [2] Claude Code security docs ↗ [3] Codex sandbox ↗ [4] Claude Code sandboxing ↗ [5] OWASP, Prompt Injection ↗

Illustrative configuration, not product defaults or a tested demo. Vendor documentation [1–4] checked October 21, 2026 (slide guide). Prompt injection: OWASP definition and illustrative scenario [5].

Try one new way to use AI this semester.

Pick one route and try it on something small, with public or synthetic data.

COMPARE, THEN SCRIPT IT

Purdue GenAI Studio

Send one prompt to two models side by side. Then send it from a short Python script through the API.

Log in: genai.rcac.purdue.edu ↗ **API guide ↗**

Approved for public and sensitive (L2) data, not restricted data such as FERPA or HIPAA records. **[1]** Keep your API key out of your code.

SEE WHERE IT RUNS

A model on your computer

Run a small open-weight model with llama.cpp or Ollama; either can serve it to your scripts.

llama.cpp ↗ **Ollama ↗**

Choose a model your hardware can run, and check whether your setup sends data anywhere else.

BUILD, THEN REVIEW

A coding agent

Try Claude Code or Codex on your own project, with approvals on. Read every change and run the tests.

How agent approvals work ↗

For coursework, follow the course's rules. Test any tool like a statistician: checkable questions, repeated runs, recorded settings.

WHERE TO GO NEXT **Now:** CS/SCI 173, the College of Science AI Competency course **[5]** **Later:** Chapter 6, LLMs in data science **[6]**

Learn one new way to use AI this semester, and keep the thinking yours.

Record the model, date, settings, and your checks so someone else can repeat the work.

Sources: [1] Purdue RCAC, GenAI Studio (Sept. 29) ↗ [2] Purdue RCAC, API guide ↗ [3] llama.cpp ↗ [4] Codex approvals & security ↗ [5] Purdue Science, CS/SCI 173 ↗ [6] Computational Data Science, Ch. 6 ↗

RCAC documentation updated September 29, 2026; check current access, limits, and data rules before you start.

Build something people can use. Learn as it grows.

SPEAKER'S SUGGESTION · A PROPOSED PROJECT, NOT A BUILT OR TESTED SYSTEM

"I want to build a platform around something I care about, put it online, and send my friends a link."

Baseball strategy platform · release 2 — Share link ↗

Explore players, build lineups, predict matchups, simulate seasons, and save and share scenarios with friends.

Data system
collect, update, and check the data; keep it in a database
databases, data checks

Documented Statcast fields: batter, pitcher, pitch type, speed, result [1]

Explore players
profiles, comparisons, trends
vectors, similarity, multivariate methods

CURRENT FOCUS
Forecast matchups
predictions with uncertainty
conditional probability, regression, regularization

Simulate seasons
games and seasons under your assumptions
Monte Carlo simulation, dependence

Optimize lineups
search lineups under constraints
combinatorics, constrained optimization

Friends open your link
They try lineups, compare saved scenarios, and tell you what is confusing, slow, or missing.

Hosted application saved scenarios, shareable links, tests; it connects every part
databases, application design, hosting

Start: use AI to help build and share a first working version.

Example cycle (proposed): forecasts swing to extremes for players with little data → learn partial pooling, combining sparse player data with broader evidence → revise, check against later games, and feed the simulator.

New course topic? Ask:
"Where could I use this in my platform?"

1 **USE AND GET FEEDBACK**
What do you and your friends want? What is confusing, unreliable, slow, or missing?

2 **CONNECT AN IMPROVEMENT TO IDEAS**
Which part needs to change? Which course topics connect? What else would help?

3 **LEARN AND EXPERIMENT**
Courses, independent resources, documentation, and AI: investigate and try alternatives.

4 **IMPROVE AND RELEASE**
Apply what you learned, check the result, and publish the next version.

↩ Each release brings new possibilities, feedback, and questions.

Outcome: a more useful platform, and you can explain its behavior.

Let the project give you reasons to learn—and use what you learn to make it better.

A personal project can use a lot of AI help; coursework follows each course's rules.

Sources: [1] Baseball Savant CSV fields ↗

A proposed project and suggested learning strategy, not a built or tested system or a validated intervention. Data fields: Baseball Savant's Statcast CSV documentation [1].

You do the thinking; let AI support your growth at Purdue.

MAIN POINTS · THE FOUR QUESTIONS FROM THE START

Purpose

What is the purpose?

- Match AI use to what you need to learn.
- Write your own connection from courses to goals.

Permission

What is allowed?

- Follow course rules now, employer policy later.
- Share data only where a tool's terms allow.

Reasoning

What must I do myself?

- The thinking: try first, then get help with the gap.
- Build the expertise to check and steer AI.

Checking

How will I check it?

- The same data can lead AI to different conclusions.
- Read the passage, check the query, review every change.

THE THINKING IS YOURS · SPEAKER'S ADVICE

Give a hard proof or project days, not minutes.

Struggling with it is how you develop. Don't be quick to turn to AI for the quick fix.

Across 53 studies, trying problems before being taught built deeper understanding. **[1]** AI users whose homework time held up showed small exam losses; larger declines went with very short homework times. **[2]**

START THIS WEEK

Pick one hard problem from your coursework and check what help is allowed. Stay with it over several days, then bring one precise question to an instructor or TA.

Questions?

Sources: [1] Sinha & Kapur (2021) ↗ [2] Strömberg et al. (2026) ↗

Main points answer the opening questions; their evidence is on each point's slide. Days, not minutes, is the speaker's advice; [1] and [2] repeat findings from slides 9 and 6.